

Gain in own-answer acc@1
from end of thought ↑

REASONING SETTING

● OpenThinker3-7B

◆ Qwen3-8B

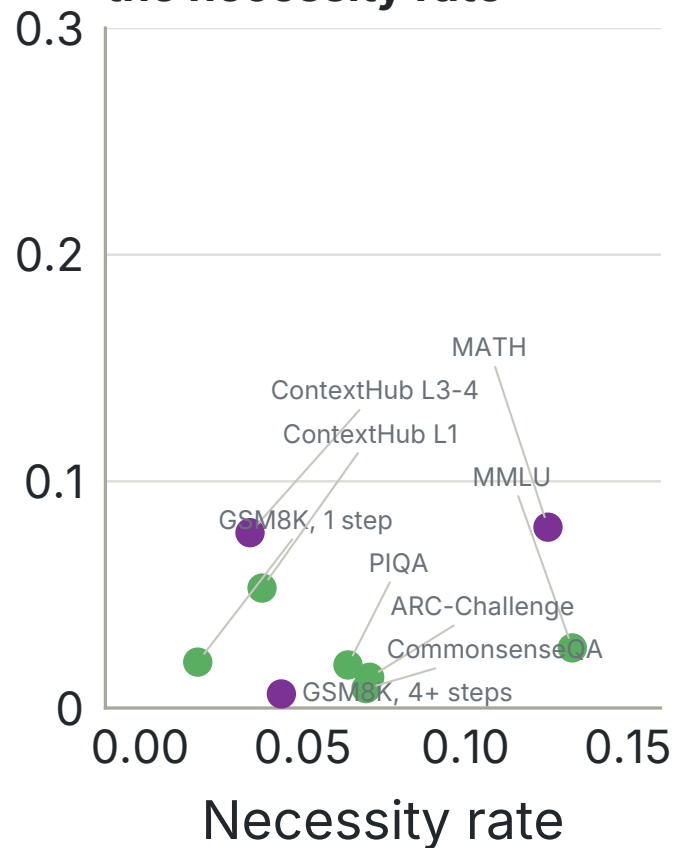
CORPORA (A, B)

■ Needs-reasoning

■ No-reasoning

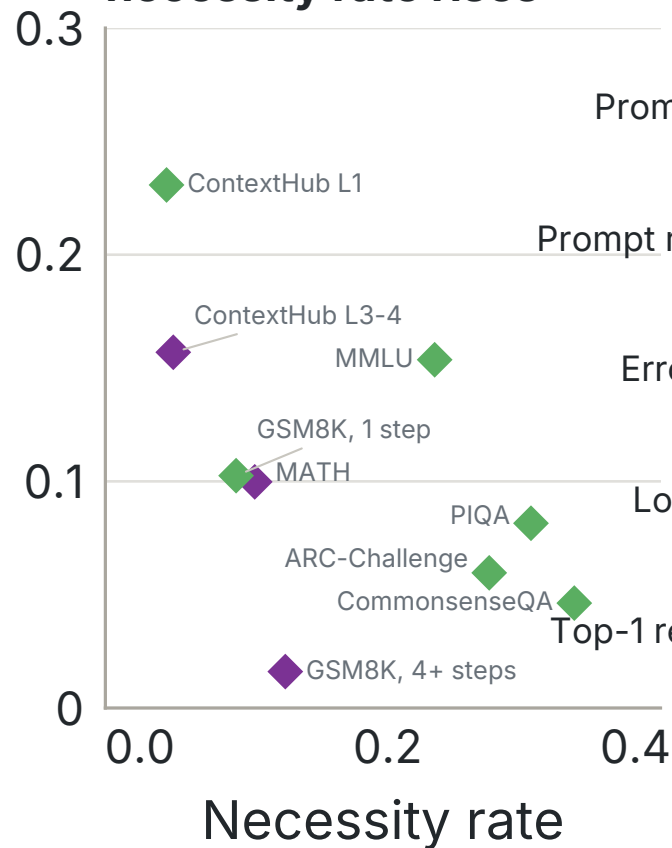
A · OPENTHINKER3-7B VS PARENT

**End-of-thought gains are small
and do not track
the necessity rate**



B · QWEN3-8B, THINKING ON VS OFF

**Larger gains for the thinking
toggle, falling as the
necessity rate rises**



C · PER-QUESTION AUROC

**No map score identifies which
MATH questions
need reasoning**

